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PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

KIBAHA TOWN COUNCIL

FORM FOUR PRE MOCK EXAMINATION

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O31 PHYSICS

Time 3:00 hours **Year:2024**

Instructions.

1. This paper consists of three section A,B and C.
2. Answer **all** questions from section A and B only, **TWO** (2) questions from section C.
3. Write your **Name** on every page of your answer sheets provided.
4. wherever necessary use the following constant
   i. Speed of light C = 3.0×10^8m/s
   ii. Speed of sound in air = 340m/s
   iii. Acceleration due to gravity, g = 10m/s^2
   iv. π = 3.14.
   v. Density of water = 1000kg/m^3
   vi. Density of kerosene= 0.920g/cm^3
   vii. The coefficient of aluminium-steel surface static friction= 0.7.
   viii. The coefficient of aluminium-steel surface kinetic friction= 0.5.

MsomiBora.com

SECTION A (16 MARKS)

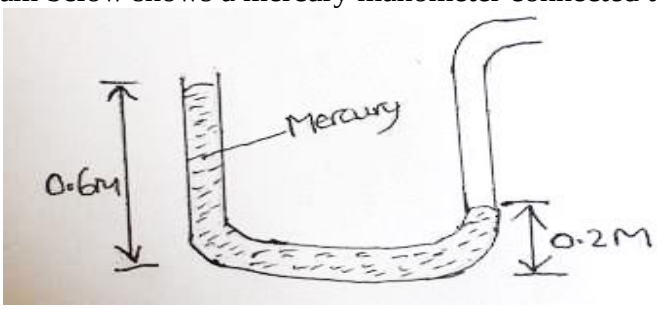
Answer all questions

1. For each of the item (i) – (x), choose the correct answer among the given alternatives.

i. Which lists contains only scalar quantities.

A. Acceleration, displacement, velocity
B. Distance, Force, speed
C. Force, length, Time
D. Length, Mass, speed
E. Distance, momentum, velocity

ii. The diagram below shows a mercury manometer connected to a gas container and tapeopened



The pressure difference between the gas and atmosphere is ?

A. 28,000Pa
B. 4200Pa
C. 544,00Pa
D. 84,000Pa
E 28,000Pa .

iii. The use of iron cylinder round the neck of cathode ray tube in TV receiver is forproviding

A. Magnetic shielding
B. Mechanical protection
C. To give spot light
D. Conduction to electricity
E. Electric and magnetic field

iv. Which of the following burning medium requires carbon dioxide extinguisher?

A. Organic solids
B. Flammable liquids and greases
C. Flammable gases
D. Combustible metals
E. Electrical hazards

v. The temperature scale where 273 represents freezing and 373 represents boiling pointsof water?

A. Fahrenheit
B. Celsius
C. Kelvin
D. Absolute
E. Anomalous.

vi. Classify the force experienced when a metal solid or hard object is twisted

A. Compression B. Stretching C. Restoring D. Torsion E. Friction

vii. Which of the following is correct about surface tension of liquids?

A. Surface tension increase with increase in temperature
B. Surface tension decrease with increase in temperature
C. Surface tension increase with addition of impurities
D. Surface tension increase due to adhesive force
E. Surface tension is not affected by temperature.

viii. The initial volume of a liquid in a burette was $57cm^3$ .If xcm^3 of liquid was poured outof the burette and final volume of the liquid remaining was $23cm^3$, The value of x was

A. 80 cm^3 B. 50 cm^3 C. 34 cm^3 D. 46 cm^3 E. -46 cm^3

ix. A typist uses a new carbon paper under her top typing paper for making a copy of a letter when she holds the carbon paper close to a plane mirror, she can read the letter this is because the mirror.

A. Forms an image the same size as object
B. Produces an inverted image
C. Produces a laterally image
D. Produces a virtual image
E. Forms an image behind the mirror.

x. A bus carrying a very big load on its carries can easily overturn because,

A. it cannot run fast
B. its equilibrium is stable
C. Its center of gravity is low
D. It is more stable and can run fast
E. Its center of gravity is high.

2. Match the items in **list A** and in **list B**

LIST A	**LIST B**
i. Fossil fuel that is dangerous as it is explosive in nature ii. Dangerous energy that emits harmful radiation and it is usuallylocated far away from human settlements iii. Energy from the sun and it is always dependent on weather condition? iv. Energy resources by which their formation takes a long time of period. v. Sustainable energy resources. vi. Good examples of non-renewable resources	A. Non-renewable energy resources. B. Solar energy C. Renewable energyresources. D. Natural gas. E. E. Wind and solarenergy. F. Coal and oil. G. Nuclear energy. H. Radioactive energy.

SECTION B (54 Marks)

Answer all questions

3. (a) A concave mirror can be used as a magnifying mirror by using a well ray diagramshow how you can obtain a magnified and virtual image of a pain object by using a concave mirror. **(4marks)**

   (b) In a certain murder investigation, it was important to discover whether the victim waslong sighted or short sighted. How could a detective decide by examining the spectacles? **(05marks)**

4. a) A boy and a girl wanted to cook same amount of beans, A boy used normal pot while girl used pressure cooker .It was observed that pressure cooker lakes short time while normal pot takes longer time to cook Explain Why ?

   (b) With the aid of relevant diagram give reason why lake water in very cold region doesnot freeze completely into ice even if the temperature of the region is far below freezing point and give the biological significance of this behaviour.

5. (a) In manufacturing industries friction force always decreases the working efficiency of amachine. If the work input on a machine is 900J and a work against friction was 50J.

   i. Calculate the efficiency of that machine **(2 marks)**

   ii. Explain two devices that can be used so as to increase the efficiency of that machine **(2 marks)**

   (b) Two forces of 8N and 6N are acting on a body perpendicular to each other find the magnitude of their resultant. **(5 marks)**

6. (a) The half-life of radon is 56 days. If the count rate mass is 600g. How long will it takefor the element to get reduced by 76%? **(5 marks)**

   (b) Why gold leaf electroscope can be used to detect only alpha particles but not beta particles and gamma radiations?. **(4 marks)**

7. (a) A force of 200N and 20cm long spanner and 150N and 30cm long spanner were separately used to loosen the left and right front car tire nuts, respectively. Show that the 30cm long spanner will be more effective than 20cm long spanner. **(4 marks)**

   (b). Why while swimming, the swimmer pushes the water backwards with his hands?. **(5 marks)**

8. (a) You as physics student you are required to advice the community on four ways they can use to rescue the world from the earths overheating**.** **(4 marks)**

   (b) With three examples differentiate Jovian planet from terrestrial planets. Hence give reason why Saturn has low gravity than earth although is larger**.** **(5 marks)**

SECTION C (30 marks)
Answer two (2) questions

9. (a) With the aid of a well labeled diagram, use the convection currents in water to explain how heat transfer by convection takes place**.** **(5 marks)**

(b) A container holds 1.5 kg of ice initially at -10°C. Heat is supplied to the container at the rate of 12.6 KJ per second.

(i) Plot a graph of temperature versus time for the 100 seconds during which heat is supplied**. (4 marks)**

(ii) What will the temperature of the contents of the container be at the end of 42.5 seconds? **(3 marks)**

(iii) Calculate the total energy to obtain water at 100°C **(3 marks)**

10. (a) By using knowledge of vector applications describe if pressure is a vector or scalar,give concrete reason

(b) Semiconductor are regarded as non-ohmic conductor while metal such as copper andsilver are regarded as ohmic conductor

i. Explain why their called so and,
ii. Distinguish between conductor and semi-conductor interm of their energyband

(c) why is it necessary for cathode ray tube to be evacuated?

11. (a) Electrical energy is distributed in all parts of Tanzania by national grid systemwhich transmits AC at very high voltage

i. Explain why the very high voltage is necessary
ii. Why is an ac and not dc used

(b) Differentiate between alpha(α), beta (β) and gamma γ radiations in magnetic field.

(c) We cannot hear echo in small room, explain why?